

THE HONG KONG POLYTECHNIC UNIVERSITY
DEPARTMENT OF APPLIED MATHEMATICS
AMA3640 STATISTICAL INFERENCE

Midterm Test Session 1

17:30–18:15, 7 Nov 2025

1. Consider a random sample $X_1, \dots, X_n$ with density

$$f(x; \theta) = 2\theta x e^{-\theta x^2} \quad \text{for } x > 0,$$

where $\theta > 0$.

- (a) Find the maximum likelihood estimator (MLE) of $1/\theta$. **[8 marks]**

- (b) Find the expected value and variance of the MLE of $1/\theta$ you found in (a). You can use the facts that

$$\int_0^\infty 2\theta x^3 e^{-\theta x^2} dx = \frac{1}{\theta}$$

and

$$\int_0^\infty 2\theta x^5 e^{-\theta x^2} dx = \frac{2}{\theta^2}.$$

[8 marks]

- (c) Find the Cramér–Rao lower bound for the variance of unbiased estimators of $1/\theta$. Does the MLE attain the lower bound? **[8 marks]**

- (d) Argue that the MLE of $1/\theta$ is consistent. **[6 marks]**

2. Consider a random sample $X_1, \dots, X_n$ with density

$$f(x; \theta) = 2e^{-2(x-\theta)} \quad \text{for } x \geq \theta,$$

where $\theta \in \mathbb{R}$.

- (a) Find the method of moments estimator of θ . **[7 marks]**

- (b) Find the MLE of θ . **[6 marks]**

- (c) Show that the MLE of θ is consistent. **[7 marks]**

***** END OF PAPER *****

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Midterm Test Session 2

17:30–18:15, 14 Nov 2025

1. Consider a random sample of four Bernoulli random variables, $X_1, \dots, X_4$, with

$$P(X_i = k) = \begin{cases} \theta & \text{for } k = 1 \\ 1 - \theta & \text{for } k = 0 \end{cases}$$

for $i = 1, \dots, 4$, where $0 < \theta < 1$.

- (a) Argue that $T \equiv \sum_{i=1}^4 X_i$ is a complete sufficient statistic for θ . **[10 marks]**

- (b) Let $\tau(\theta) = E\{X_1 I(X_2 < X_3 + X_4)\} = 2\theta^4 - 3\theta^3 + 2\theta^2$. Clearly,

$$\hat{\tau} \equiv X_1 I(X_2 < X_3 + X_4)$$

is an unbiased estimator of $\tau(\theta)$. Find the best unbiased estimator of $\tau(\theta)$ by taking conditional expectation of $\hat{\tau}$ given T . Hint: Given $T = t$ ($t = 0, 1, 2, 3, 4$), all possible arrangements of (X_1, X_2, X_3, X_4) with exactly t ones and $(4 - t)$ zeros are equally likely. **[15 marks]**

2. Consider a random sample $X_1, \dots, X_n$ with density

$$f(x; \alpha, \beta) = \frac{\alpha x^{\alpha-1}}{\beta^\alpha} \quad \text{for } 0 \leq x \leq \beta,$$

where $\alpha > 0$ and $\beta > 0$.

- (a) Find a two-dimensional sufficient statistic for (α, β) . **[8 marks]**
- (b) Find the maximum likelihood estimators (MLE) of α and β . Hint: find the MLE of β first, which does not depend on the value of α . **[10 marks]**
- (c) Suppose it is known that $\alpha = 2$ and $\max_{i=1, \dots, n} X_i$ is a complete sufficient statistic for β . Find the best unbiased estimator of β^2 . Justify that your answer is indeed the best unbiased estimator. You can use the fact that $E(\max_{i=1, \dots, n} X_i^2) = n\beta^2/(n+1)$. **[7 marks]**

***** END OF PAPER *****

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Midterm Test Session 1 Solution

1. (a) The likelihood function is

$$L(\theta) = \prod_{i=1}^n 2\theta X_i e^{-\theta X_i^2} = 2^n \theta^n \left(\prod_{i=1}^n X_i \right) e^{-\theta \sum_{i=1}^n X_i^2}.$$

The log-likelihood is

$$\log L(\theta) = n \log 2 + n \log \theta + \sum_{i=1}^n \log X_i - \theta \sum_{i=1}^n X_i^2.$$

Differentiating and setting to zero,

$$\frac{\partial}{\partial \theta} \log L(\theta) = \frac{n}{\theta} - \sum_{i=1}^n X_i^2 = 0 \implies \hat{\theta} = \frac{n}{\sum_{i=1}^n X_i^2}.$$

The second derivative is $-\frac{n}{\theta^2} < 0$, so $\hat{\theta}$ is indeed a maximizer.

The MLE of $1/\theta$ is

$$\frac{1}{\hat{\theta}} = \frac{1}{\bar{\theta}} = \frac{1}{n} \sum_{i=1}^n X_i^2.$$

- (b) We have

$$E(X_1^2) = \int_0^\infty 2\theta x^3 e^{-\theta x^2} dx = \frac{1}{\theta}.$$

Thus,

$$E\left(\frac{1}{\bar{\theta}}\right) = \frac{1}{n} \sum_{i=1}^n E(X_i^2) = \frac{1}{\theta}.$$

For the variance,

$$\text{Var}(X_1^2) = E(X_1^4) - E(X_1^2)^2 = \int_0^\infty 2\theta x^5 e^{-\theta x^2} dx - \left(\frac{1}{\theta}\right)^2 = \frac{2}{\theta^2} - \frac{1}{\theta^2} = \frac{1}{\theta^2}.$$

Therefore,

$$\text{Var}\left(\frac{1}{\bar{\theta}}\right) = \frac{1}{n} \text{Var}(X_1^2) = \frac{1}{n\theta^2}.$$

- (c) The Fisher information for one observation is

$$I(\theta) = E\left[-\frac{\partial^2}{\partial \theta^2} \log f(X_1; \theta)\right] = \frac{1}{\theta^2}.$$

For n observations, the information is n/θ^2 .

The Cramér–Rao lower bound for unbiased estimators of $g(\theta) = 1/\theta$ is

$$\frac{[g'(\theta)]^2}{nI(\theta)} = \frac{(-1/\theta^2)^2}{n/\theta^2} = \frac{1}{n\theta^2}.$$

The MLE attains the lower bound.

(d) Since $\frac{1}{\hat{\theta}}$ is unbiased and

$$\text{Var}\left(\frac{1}{\hat{\theta}}\right) = \frac{1}{n\theta^2} \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

it follows that $\frac{1}{\hat{\theta}}$ is consistent for $1/\theta$.

2. (a) The expected value is

$$E(X) = \int_{\theta}^{\infty} 2xe^{-2(x-\theta)}dx = \int_0^{\infty} 2(y+\theta)e^{-2y}dy = \int_0^{\infty} 2ye^{-2y}dy + \theta \int_0^{\infty} 2e^{-2y}dy = 1 + \theta.$$

The method of moments estimator is

$$\tilde{\theta} = \bar{X} - 1.$$

(b) The likelihood is

$$L(\theta) = \prod_{i=1}^n 2e^{-2(X_i-\theta)}I(X_i \geq \theta) = 2^n e^{-2(\sum X_i - n\theta)}I(\theta \leq \min_i X_i).$$

The part outside the indicator function is increasing in θ . Therefore, the likelihood is maximized when θ is as large as possible, i.e.,

$$\hat{\theta} = \min_i X_i.$$

(c) For any $\varepsilon > 0$,

$$\begin{aligned} P(|\min_i X_i - \theta| > \varepsilon) &= P(\min_i X_i > \theta + \varepsilon) \\ &= [P(X_1 > \theta + \varepsilon)]^n \\ &= \left(\int_{\theta+\varepsilon}^{\infty} 2e^{-2(x-\theta)}dx \right)^n \\ &= (e^{-2\varepsilon})^n = e^{-2n\varepsilon} \rightarrow 0 \quad \text{as } n \rightarrow \infty. \end{aligned}$$

Therefore, $\min_i X_i$ is consistent for θ .

Midterm Test Session 2 Solution

1. (a) The pmf of X_i can be written as

$$f(x) = \theta^x(1-\theta)^{1-x} = (1-\theta) \left(\frac{\theta}{1-\theta} \right)^x = (1-\theta)e^{x \log(\theta/(1-\theta))}.$$

Therefore, the distribution belongs to the exponential family. We conclude that $T = \sum_{i=1}^4 X_i$ is sufficient for θ . Also, since $\{\theta \in (0, 1) : \log(\theta/(1-\theta))\}$ contains an open interval, T is also complete.

- (b) Consider

$$E(X_1 I(X_2 < X_3 + X_4) \mid T = t).$$

For $t = 0$, $X_1 = X_2 = X_3 = X_4 = 0$, so $X_1 I(X_2 < X_3 + X_4) = 0$ with probability 1.

For $t = 1$, either $X_1 = 0$ or $X_2 = X_3 = X_4 = 0$, so $X_1 I(X_2 < X_3 + X_4) = 0$ with probability 1.

For $t = 2$, $X_1 I(X_2 < X_3 + X_4) = 1$ if $X_1 = 1$ and X_3 or $X_4 = 1$. Thus,

$$\begin{aligned} E(X_1 I(X_2 < X_3 + X_4) \mid T = 2) &= P(X_1 = X_3 = 1, X_2 = X_4 = 0 \mid T = 2) \\ &\quad + P(X_1 = X_4 = 1, X_2 = X_3 = 0 \mid T = 2) \\ &= \frac{1}{3}. \end{aligned}$$

For $t = 3$,

$$E(X_1 I(X_2 < X_3 + X_4) \mid T = 3) = P(X_1 = X_3 = X_4 = 1, X_2 = 0 \mid T = 3) = \frac{1}{4}.$$

For $t = 4$, $X_1 I(X_2 < X_3 + X_4) = 1$ with probability 1.

The best unbiased estimator of τ is

$$E(X_1 I(X_2 < X_3 + X_4) \mid T) = \begin{cases} 0 & \text{for } T = 0, 1, \\ \frac{1}{3} & \text{for } T = 2, \\ \frac{1}{4} & \text{for } T = 3, \\ 1 & \text{for } T = 4. \end{cases}$$

2. (a) The likelihood is

$$L(\alpha, \beta) = \prod_{i=1}^n \alpha X_i^{\alpha-1} \beta^{-\alpha} I(X_i \leq \beta) = \alpha^n \left(\prod_{i=1}^n X_i \right)^{\alpha-1} \beta^{-n\alpha} I(\max X_i \leq \beta).$$

By the factorization theorem, $(\prod_{i=1}^n X_i, \max X_i)$ is a sufficient statistic for (α, β) .

- (b) Note that for any $\alpha > 0$, $\beta^{-n\alpha}$ is decreasing in β . Therefore, the MLE of β is

$$\hat{\beta} = \max_i X_i.$$

The log-likelihood at $\beta = \hat{\beta}$ is

$$\log L(\alpha, \hat{\beta}) = n \log \alpha + (\alpha - 1) \sum_{i=1}^n \log X_i - n\alpha \log \hat{\beta}.$$

Differentiating with respect to α and setting to zero,

$$\frac{\partial}{\partial \alpha} \log L(\alpha, \hat{\beta}) = \frac{n}{\alpha} + \sum_{i=1}^n \log X_i - n \log \hat{\beta} = 0 \implies \hat{\alpha} = \frac{n}{n \log \hat{\beta} - \sum_{i=1}^n \log X_i}.$$

The second derivative is $-\frac{n}{\alpha^2} < 0$, so $\hat{\alpha}$ is the maximizer.

(c) Since $E(\max_i X_i^2) = \frac{n\beta^2}{n+1}$,

$$\frac{n+1}{n} (\max_i X_i)^2$$

is an unbiased estimator for β^2 . Because this is a function of a complete sufficient statistic, by the Lehmann–Scheffé theorem, this is the best unbiased estimator for β^2 .